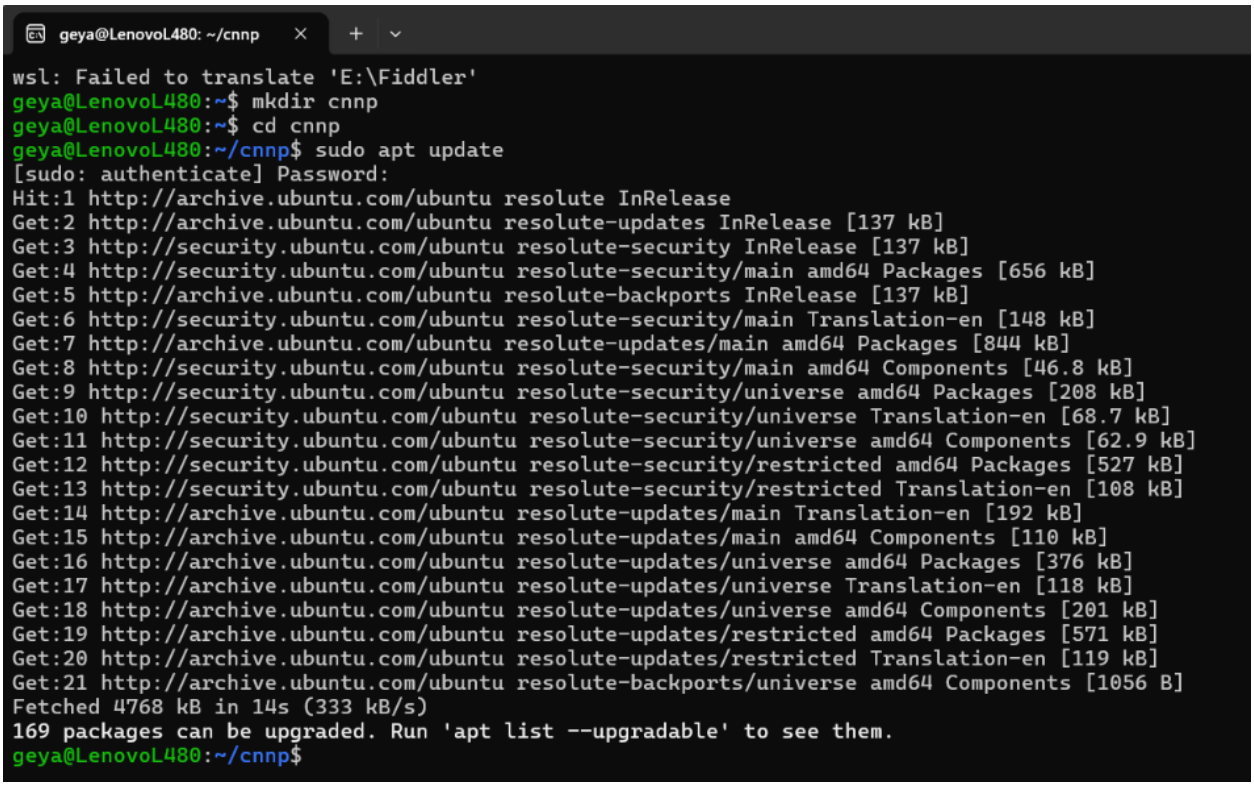
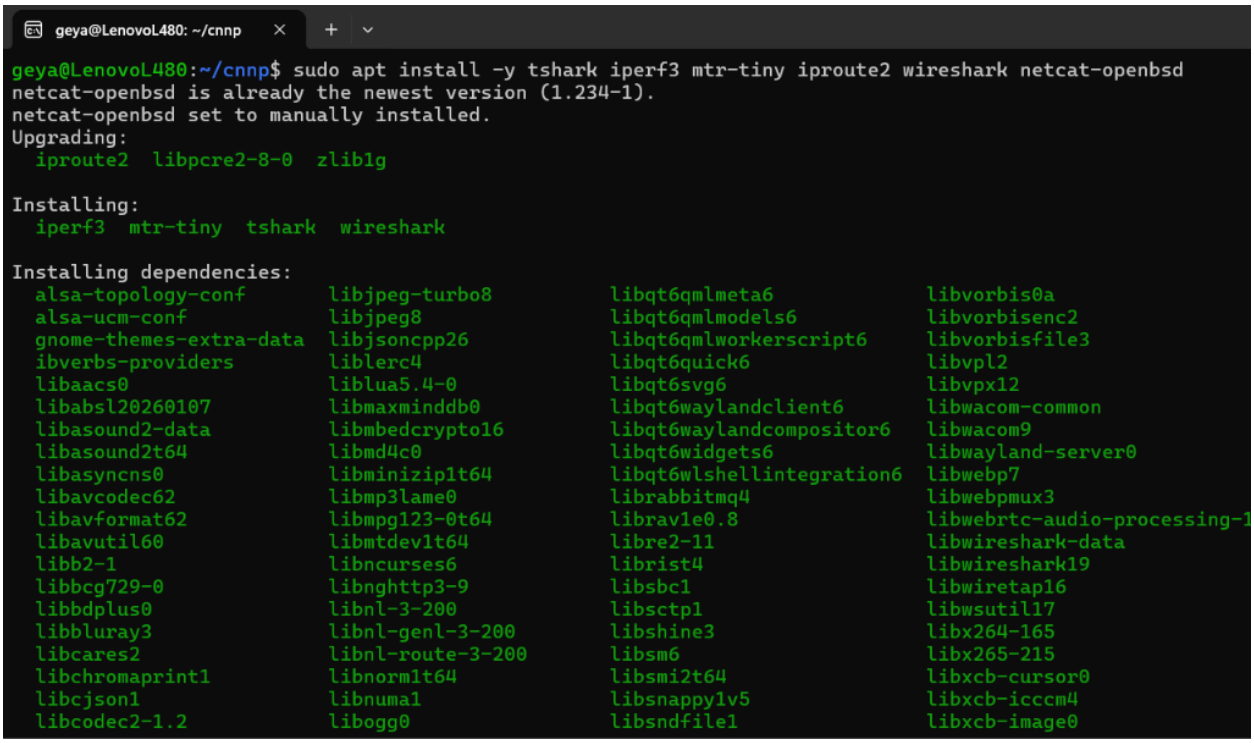
Skill -2

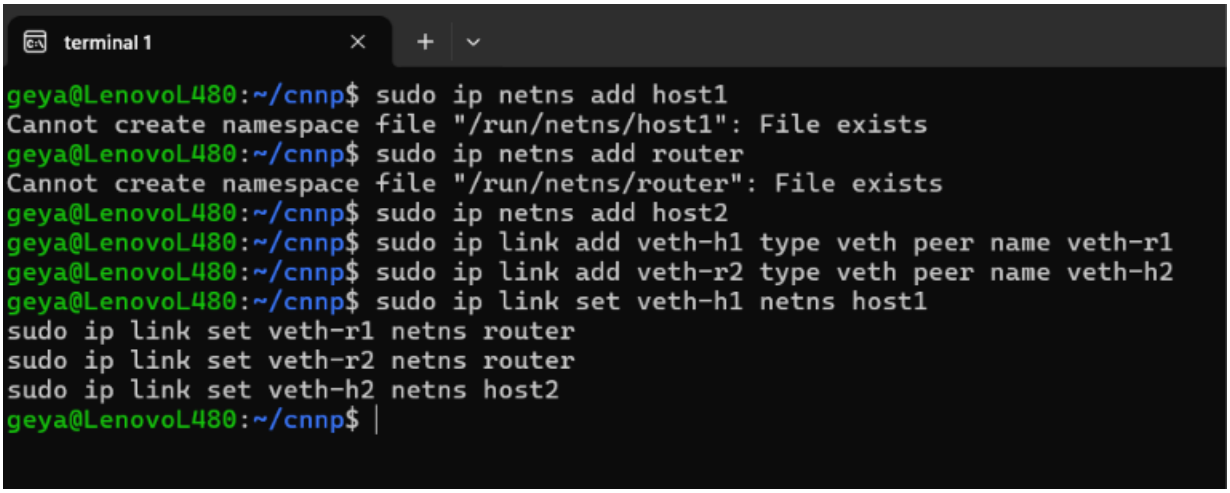
2520090123 S-11

Step 1: Install Tools



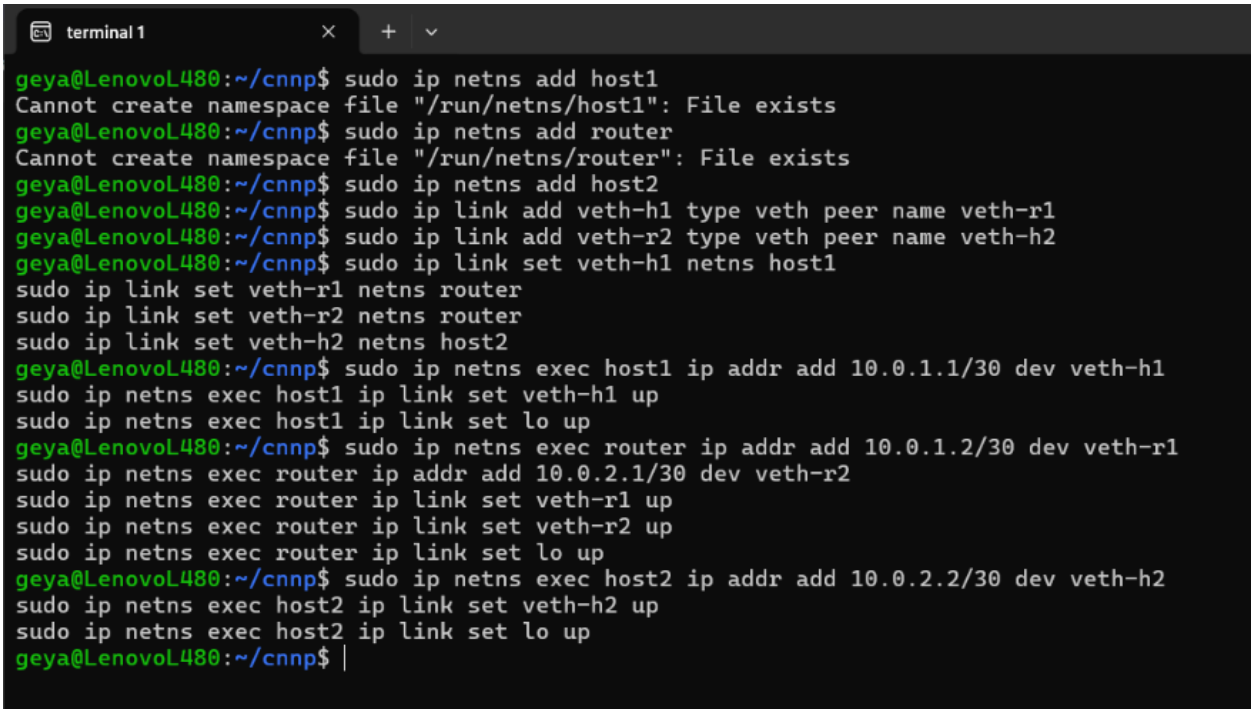


Step 2: Build the network

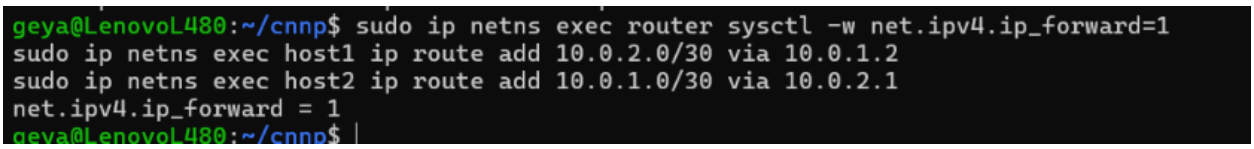


Step 3: Configure IP address

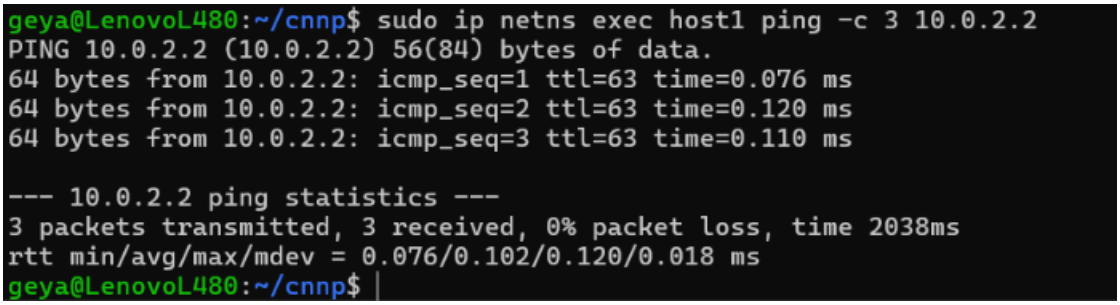
host1, router, host2



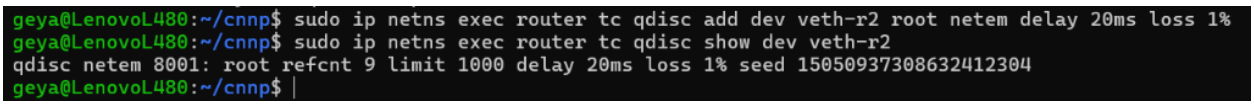
Step 4: Enable routing



Step 5: Test Connectivity

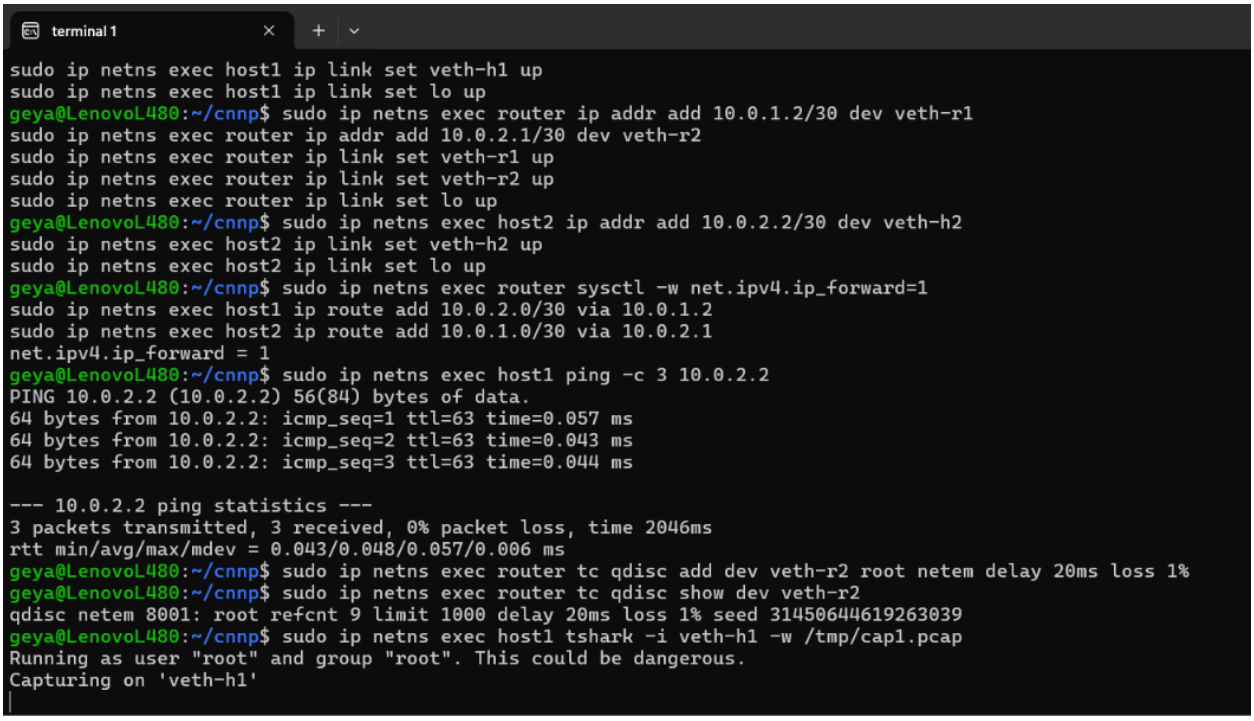


Step 6: Add delay and loss

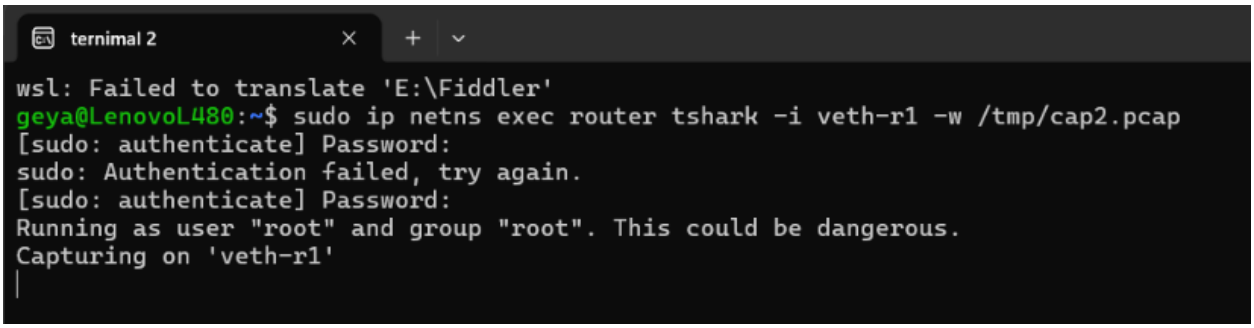


Step 7,8: Open 4 terminals

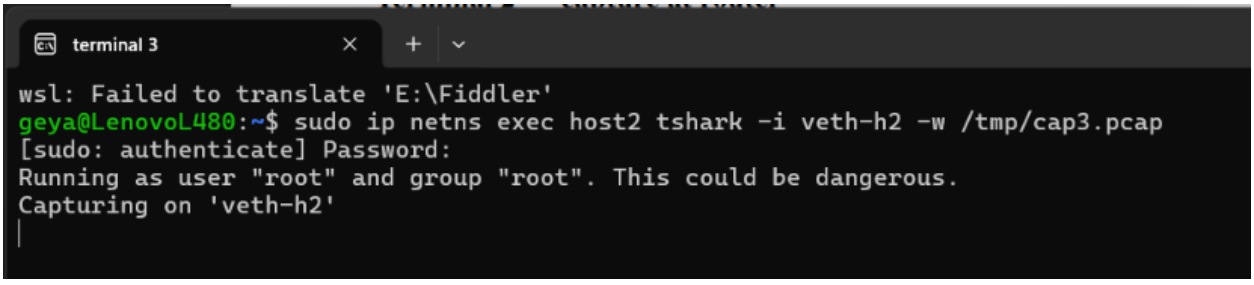
Terminal 1-



Terminal 2-

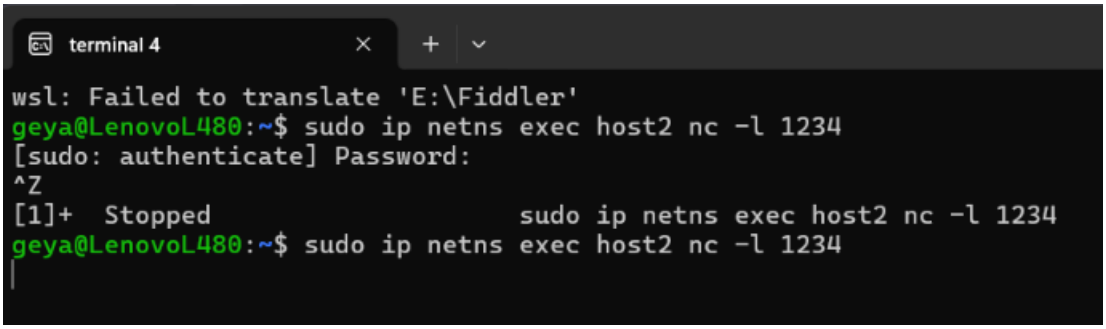


Terminal 3-

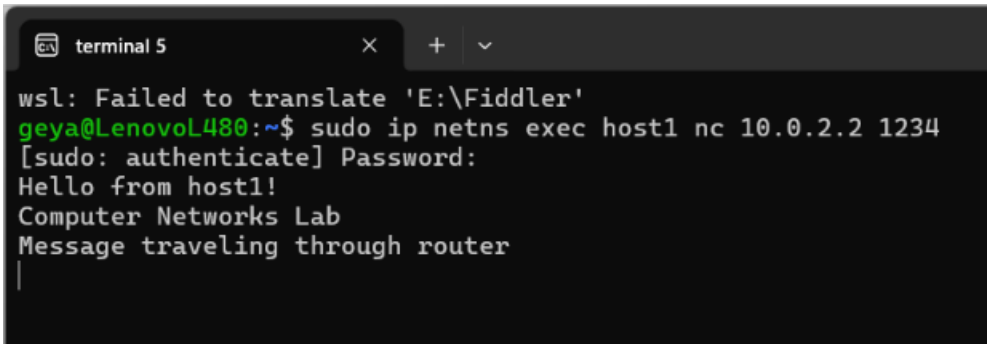


Step 9: Send real message from host1 to host2

Receiver on host2

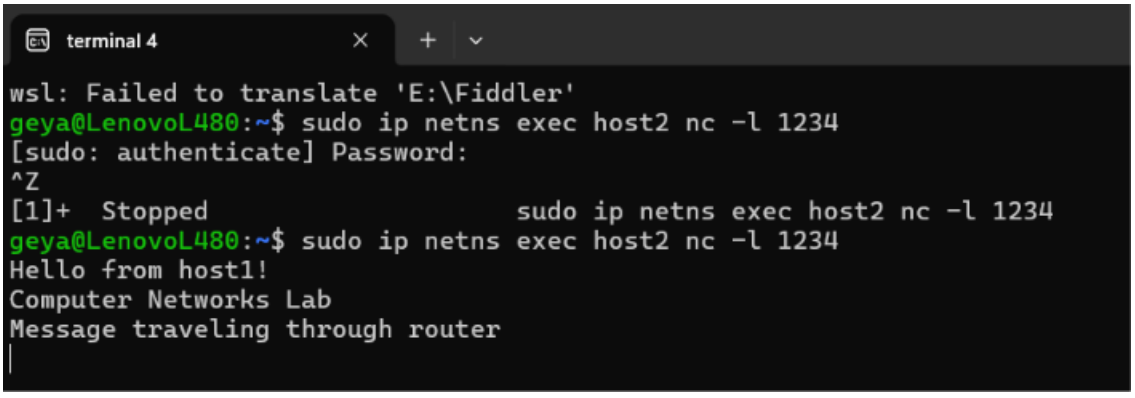


Terminal 5-



Step 10: What you see in real time

Terminal 4 (host2) shows your message arriving:



Terminal 1,2,3 count packets

```
geya@LenovoL480:~/cnnp$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
34
```

```
ternimal 2
wsl: Failed to translate 'E:\Fiddler'
geya@LenovoL480:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
34
```

```
terminal 3
wsl: Failed to translate 'E:\Fiddler'
geya@LenovoL480:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
36
```

STAGE 12 — Fix permissions

```
terminal 4
geya@LenovoL480:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
[sudo: authenticate] Password:
geya@LenovoL480:~$
```

STAGE 13 — Check TTL

```
terminal 1
geya@LenovoL480:~/cnnp$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     10 63
     12 64
geya@LenovoL480:~/cnnp$
```

```
ternimal 2
geya@LenovoL480:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     10 63
     12 64
geya@LenovoL480:~$
```

```
terminal 3
geya@LenovoL480:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     11 63
     10 64
geya@LenovoL480:~$
```

```
terminal 4
[2]-  Exit 126                 wireshark /tmp/cap1.pcap
geya@LenovoL480:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 3.9K Nov  9 17:15 /tmp/cap1.pcap
-rw-r--r-- 1 root root 3.9K Nov  9 17:15 /tmp/cap2.pcap
-rw-r--r-- 1 root root 4.1K Nov  9 17:16 /tmp/cap3.pcap
geya@LenovoL480:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     10 63
     12 64
geya@LenovoL480:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     10 63
     12 64
geya@LenovoL480:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     11 63
     10 64
geya@LenovoL480:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
exec of "mtr" failed: Input/output error
geya@LenovoL480:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative
250.577944376
371.005337953
geya@LenovoL480:~$ tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative
250.577940876
371.005333653
geya@LenovoL480:~$ tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative
203.473978573
323.901384150
geya@LenovoL480:~$ sudo ip netns del host1
sudo ip netns del router
sudo ip netns del host2
geya@LenovoL480:~$ sudo ip netns list
geya@LenovoL480:~$
```

3 point capture complete